RansKnow: An Open Dataset of Ransomware-Related YouTube Transcripts with MITRE ATT&CK Annotations

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Abstract

We present RansKnow, an open corpus of 1,128 ransomware-related YouTube video records (1,034 with extracted transcripts) drawn from 92 cybersecurity channels, with MITRE ATT&CK-aligned structured features – ransomware family, tactic, tool, and platform mentions – produced by a deterministic regex-based extraction pipeline. This paper's contribution is not the corpus alone: a systematic label-quality audit found that 9 of 41 ransomware-family aliases collide with ordinary English or business words (e.g. `play`, `hive`, `cactus`) badly enough to inflate the single most-detected family by roughly $3.5\times$ before a context-based disambiguation fix, and a counterfactual-explanation analysis later in the paper shows a trained classifier partially inherited that same word-level fragility even after the fix. We state plainly here, not only in a limitations section, that this project does not include an independently human-annotated evaluation set: every baseline, calibration, and conformal-prediction result we report is agreement with this audited-but-imperfect extraction pipeline, not verified real-world accuracy, and we frame every such number accordingly throughout. Under that scope, weak-label baselines reach macro-F1 0.90 on a binary relevance screen and 0.81 on dominant-tactic classification, but remain weak (macro-F1 $\leq$0.20) on the higher-cardinality family and tool taxonomies; calibration against the weak labels is moderate (Expected Calibration Error 0.02–0.11) with a Random Forest tree-vote-entropy signal that separates label agreement from disagreement well (AUROC up to 0.98); and split conformal prediction achieves close to its 90% target coverage with no severe conditional-coverage collapse on two out-of-calibration-distribution slices tested. Out-of-distribution detection, which does not depend on label correctness, tells a more cautionary story: standard softmax- confidence-based methods perform near chance across four realistic distribution-shift setups, including a diagnosed domain-specific failure mode where the classifier's most confident predictions coincide with exactly the signature that off-topic input also produces; only embedding-based Mahalanobis distance clears chance, and only on one of the four setups. We release the corpus, the full extraction and audit pipeline, and all experiment code openly, and argue that the audit methodology – not just its findings – is reusable by any weak-label security-NLP dataset built the same way.

Preprint.

1 Introduction

Ransomware knowledge is scattered across the same kind of source material that generated it in the first place: incident-response writeups, conference talks, and vendor threat-intelligence reporting, almost none of it structured for downstream NLP or ML use. Structured resources do exist – the MITRE ATT&CK framework itself [Strom et al., 2018], and datasets built against it from written CTI text (Section 2) – but no open, transcript-level corpus covers the specific and large body of ransomware knowledge that security professionals communicate *by speaking*: conference talks, vendor webinars, and independent-researcher walkthroughs published as video. YouTube specifically is a useful aggregation point for this material precisely because of its source diversity: DFIR vendors publishing incident retrospectives, security conferences publishing full talk recordings, and independent researchers publishing technical walkthroughs all use the same platform, under different incentives and house styles, in a way no single vendor blog or CTI feed replicates. That diversity is also why transcribing it is nontrivial – 92.7% of this corpus required automatic speech recognition rather than official captions (Section 3.3), which turns out to matter for downstream label quality, not just as a data-engineering footnote (Section 4.1).

This paper makes five contributions. First, RansKnow itself: 1,128 video records, 1,034 with extracted transcripts, from 92 cybersecurity YouTube channels, with MITRE ATT&CK-aligned structured features (Section 3). Second, a label-quality audit methodology and its concrete findings (Section 4): a systematic, dictionary-driven search for generic-word alias collisions in the regex-based extraction pipeline, which found that 9 of 41 family aliases collide with ordinary English or business words badly enough to inflate the single most-detected family by roughly $3.5\times$ before a disambiguation fix (Table 1). Third, an experiment framework and weak-label baseline results across five prediction tasks, three feature representations, and five models (Section 5), reported throughout – here in the introduction, not only in Section 10 where a reader could miss it – as agreement with the audited extraction pipeline, *not* as independently verified accuracy: this project does not include a human-annotated evaluation set held out from the labeling pipeline (Section 4.3), and every quantitative claim from Section 5 through Section 7 should be read with that scope in mind. Fourth, out-of-distribution robustness results (Section 8) that are explicitly *not* subject to that caveat, since they measure whether a model's own confidence or embedding behavior differs on distributionally shifted input, which is answerable without knowing whether any label is correct. Fifth, counterfactual and perturbation-robustness interpretability results (Section 9) that connect back to the audit: a trained classifier is shown to have partially inherited the same word-level fragility Finding A identifies in the labeling pipeline itself, evidence that the audit's concerns are not confined to the raw labels but propagate into what gets learned from them.

2 Related Work

Ransomware and cyber-threat-intelligence datasets. The MITRE ATT&CK framework [Strom et al., 2018] that RansKnow's tactic taxonomy is aligned to was designed for exactly this kind of cross-source mapping, and prior datasets have built structured resources against it from written cyber-threat-intelligence (CTI) text: MalwareTextDB [Lim et al., 2017] provides span-level annotations (39 APT reports, 6,819 sentences) for malware characteristics, and the more recent AnnoCTR [Lange et al., 2024] links entities, tactics, and techniques across 400 commercial CTI reports, with human annotation throughout. Both are built from written prose – vendor blog posts and reports, produced already in structured-adjacent form for a threat-intelligence-consuming audience. RansKnow's source material is instead spoken narration, transcribed at 92.7% via ASR (Section 3.3), from a broader and more heterogeneous set of speakers (vendor analysts, conference presenters, independent researchers) than a CTI report's house style, and its structured features are produced entirely by an audited regex/keyword pipeline (Section 4) rather than human annotation – a different, explicitly weaker, point on the label-quality spectrum those two datasets occupy, traded for scale and openness (no comparable open, transcript-level, MITRE-aligned video corpus exists at the time of writing).

Weak and distant supervision. Using an automated heuristic in place of human labels is a well-studied tradeoff, not a shortcut unique to this project: Mintz et al. [2009] formalized distant supervision for relation extraction by aligning text against a knowledge base rather than hand-labeling it, accepting a known noise ceiling in exchange for scale, and Ratner et al. [2017]'s Snorkel generalizes the idea to multiple, possibly conflicting, weak labeling functions combined probabilistically. The

Knowledge Agent (Section 4) is a single deterministic labeling function rather than an ensemble of weak sources, which makes it simpler to audit (the alias-collision analysis in Section 4.1 would be considerably harder against a black-box combination of many weak labelers) but removes Snorkel-style noise-cancellation-via-redundancy as a mitigation — a tradeoff made explicit rather than left implicit, and the direct motivation for Section 4.3's scope note running through every downstream section.

Conformal prediction and out-of-distribution detection. Split conformal prediction [Vovk et al., 2005, Lei et al., 2018] and its distribution-free coverage guarantee (Section 7) have seen growing NLP and security application as a distribution-free alternative to ad hoc confidence thresholds; Angelopoulos and Bates [2023] survey the modern toolkit this paper's Section 7.1 draws on directly. Out-of-distribution detection (Section 8) via softmax-confidence baselines [Hendrycks and Gimpel, 2017], energy scores [Liu et al., 2020], and class-conditional embedding distance [Lee et al., 2018] is comparatively less explored specifically for weak-label security-NLP classifiers, where – as Section 8.2 shows concretely – the usual confidence-based methods can fail in a domain-specific way (a classifier's most confident predictions coinciding with exactly the "nothing detected" signature that off-topic input also produces).

3 Dataset Construction

RansKnow [Kabuye, 2026] consists of 1,128 video records sourced from 92 cybersecurity YouTube channels, of which 1,034 have an extracted transcript with real content; the remaining 94[1] could not be transcribed by any method in the fetch cascade.

3.1 Channel selection

Channels were curated in two batches. Batch 1 (50 channels, `C01`–`C50`) covers established DFIR vendors, security conferences, and independent researchers. Batch 2 (51 candidate channels, `C51`–`C101`) expanded coverage into threat-intelligence vendors, additional conferences, and OT/ICS-security and government channels; 42 of the 51 candidates yielded usable transcripts, 1 was a confirmed duplicate of an existing Batch 1 channel, 4 had no discoverable dedicated YouTube channel, and 4 resolved to real channels with zero ransomware-keyword-matching content in their upload history. Full channel-level provenance for both batches is in `rubrics/Dataset_Channel_Registry_Updated_50_fixed_urls.xlsx` (Batch 1) and `rubrics/Dataset_Channel_Registry_Batch2.xlsx` (Batch 2), released alongside the corpus rather than reproduced inline as a 92-row table.

3.2 Video selection

Within each channel, videos were selected by scanning recent uploads for a fixed 21-term keyword list (`ransomware`, `extortion`, `lockbit`, `incident response`, . . . ; full list in Appendix A) matched against title and description, capped at 20 videos per channel and a minimum duration of 5 minutes (short clips and trailers excluded). This is a convenience sample, not a random or exhaustive one, and the selection mechanism itself introduces bias worth naming rather than assuming away: a video is only in scope if its *title or description* happens to contain one of the 20 terms, which conditions the corpus on each channel operator's own titling and SEO/description-writing habits. A channel that discusses ransomware extensively but titles episodes generically (e.g. by client-engagement number rather than threat name) would be under-represented regardless of its actual content relevance, and channels that lean on a small number of high-profile named families (`lockbit`, `conti`, . . .) in titles for engagement will be over-represented relative to channels discussing less headline-grabbing groups. This compounds with Section 10's broader sampling-bias point: keyword-based inclusion is a second, independent bias layer on top of channel curation, not a neutral technical filtering step.

[1] Verified directly against the released `transcripts/` tree, not assumed from $1128 - 1034$: all 94 are confirmed failed transcript extraction (no method in the fetch cascade, Section 3.3, produced usable text), not videos excluded post-fetch for another reason.

3.3 Transcript acquisition

Transcripts were acquired via a three-stage cascade: (1) the YouTube Data API's official captions where available; (2) `yt-dlp`'s auto-generated captions; (3) local Whisper transcription as a fallback. **This cascade matters more than a footnote**: only 74 of 1,034 transcripts (7.2%) came from official captions. 959 (92.7%) are ASR output – 554 from `whisper_base` (the smallest, least accurate Whisper checkpoint) and 405 from a local Whisper fallback, 1 from `yt-dlp` captions. This raw composition is Finding B of the label-quality audit (Section 4); the downstream implication – what an ASR-dominated corpus does to jargon-term detection specifically – is reported there rather than duplicated here.

4 Label-Quality Audit

The Knowledge Agent pipeline (`Scripts/knowledge_agent.py`) extracts structured features from every transcript via regex matching: ransomware family mentions (against a maintained alias list), MITRE ATT&CK tactic mention counts across 10 tactics, offensive tool mentions across 8 tools, and platform signals (Windows/Linux/ESXi). Three concrete problems were found and are described here rather than left as an implicit caveat, because a downstream classifier trained on these labels inherits whatever bias they contain.

4.1 Finding A: generic-word alias collisions

Nine of 41 family aliases collide with common English or business words (`play`, `hive`, `cactus`, `agenda`, `clop`, `inc`, `maze`, `medusa`, `royal`) – identified programmatically via dictionary lookup against `/usr/share/dict/words`, then confirmed by manual transcript inspection (e.g. `hive` matching "registry hive," a routine DFIR term, rather than Hive ransomware). A windowed-context disambiguation fix (requiring a ransomware-context term within ∼15 tokens of the match) was applied and reduces false-positive volume substantially without eliminating it – residual false positives still occur when context vocabulary is itself dense (see `Scripts/audit_family_aliases.py` for the audit methodology and Table 1 for before/after counts).

Table 1: Before/after family mention counts for the 9 flagged aliases, windowed-context disambiguation applied. `Play` alone accounted for 386 of 494 (78%) of all family-tagged videos before the fix.

Family	Before	After	Dropped
Play	386	111	275
Qilin	88	21	67
Hive	23	10	13
Maze	11	5	6
Royal	28	19	9
INC	10	7	3
Cactus	9	6	3
Cl0p	11	11	0
Medusa	7	7	0

4.2 Finding B: ASR-dominated transcript quality

As noted in Section 3.3, 92.7% of transcripts are ASR output rather than official captions.

[TODO] Run the Phase 0.4 spot-check (manual comparison of ASR output against source audio for a fixed jargon term list – `LockBit`, `Mimikatz`, `PsExec`, `Cobalt Strike`, `rclone`, `BloodHound` – across the whisper-transcribed subset) and report a measured word-error-rate-on-jargon figure here. Aggregate detection-rate comparisons across `Transcript_Provider` did not show an obvious gap, which is not the same as no effect – individual mistranscriptions of rare jargon can matter without moving an aggregate rate.

4.3 Scope: no independent ground truth

An earlier design stage of this project planned a held-out, independently annotated evaluation subset (180 videos, stratified by channel type, year, and current family-count) specifically to break the circularity of evaluating a classifier against the same weak-label pipeline it was trained on. That subset was never annotated. External annotator recruitment required institutional ethics approval not available to this project; a single-investigator pass was considered as a fallback and rejected as still too close to the system under evaluation to serve as independent ground truth in a way worth claiming in a paper.

The consequence is stated here plainly, and repeated at the point every affected result is reported, rather than left as a single Limitations bullet a reader could miss: **every quantitative result in Section 5 onward, with the explicit exception of Section 8's out-of-distribution results, measures agreement with the audited Knowledge Agent extraction pipeline (Section 4), not verified accuracy against human judgment.** Section 4 demonstrates concretely that this pipeline's labels contain real, non-trivial error even after the disambiguation fix (Finding A); nothing downstream corrects for that. Readers should treat macro-F1 and calibration numbers in this paper as characterizing how well models reproduce a specific, imperfect, documented extraction process – not as claims about ransomware-domain understanding in an absolute sense. Section 10 returns to this as the paper's central limitation.

5 Task Definitions and Baselines

Five prediction tasks are defined over the corpus: ransomware family (multi-label, 18 classes after excluding 3 singleton-occurrence classes, see Table 2), dominant MITRE tactic (multi-class, 11 classes), platform signal (multi-class, 4 classes), offensive tool mentions (multi-label, 7 classes after excluding `MegaNZ`, which has zero occurrences in the corpus), and a ransomware-relevance binary screen (derived, not a stored column: `Family_Count > 0 OR Tactic_Impact > 0`).

Table 2: Task taxonomy.

Task	Type	Classes	Label source
Family	multi-label	18	alias regex match (audited, §4.1)
Dominant tactic	multi-class (+none)	11	max of 10 tactic-pattern counts
Platform signal	multi-class (+none)	4	max of 3 platform-pattern counts
Tool mentions	multi-label	7	per-tool regex match
Ransomware-relevant	binary	2	derived

5.1 Model ladder and validation protocol

Each task is evaluated over three feature representations (structured Knowledge Agent columns, TF-IDF, mean-pooled sentence embeddings[2]) and five models (the rule-based Knowledge Agent itself as baseline 0, Logistic Regression, Linear SVM, Random Forest, LightGBM), under three cross-validation protocols: stratified random k-fold, `GroupKFold` by `Channel_ID` (to detect channel-style leakage – a model can learn "this is CrowdStrike's narration style" instead of the content), and a temporal split (train pre-2024, test 2024+, since 79% of the corpus is from 2024 onward).

5.2 Results

Per Section 4.3, every number in this section is weak-label agreement – how well a model reproduces the Knowledge Agent's own extraction, not verified accuracy. The rule-based Knowledge Agent baseline scores a tautological macro-F1 of 1.000 on every task (checked against its own labels by construction; included as baseline 0, not as evidence of anything) and is omitted from Table 3 accordingly.

[2] 384-dim `all-MiniLM-L6-v2`, chunked at $\sim$200 words per chunk and mean-pooled, computed via a dedicated Python environment (`torch` 2.5.1) since the reference implementation's default environment ships an older `torch` that cannot satisfy `sentence_transformers`' requirements; see `Scripts/precompute_embeddings.py`.

Table 3: Best stratified-CV macro-F1 per task, across every representation (structured / TF-IDF / embedding) and trained model (LogReg, Linear SVM, Random Forest, GBM) tried. Weak-label agreement, not verified accuracy (§4.3).

Task	Best representation	Best model	Macro-F1
Ransomware-relevant	TF-IDF	GBM	0.903
Dominant tactic	structured	GBM	0.813
Platform signal	TF-IDF	GBM	0.606
Tool mentions	TF-IDF	LogReg	0.200
Family	embedding	LogReg	0.129

Two patterns worth naming, both consistent with Section 9's independent checks: (1) tree models (GBM in particular) are the strongest model wherever tested, and structured features remain the strongest representation specifically for `dominant_tactic` – unsurprising, since those features are numerically close to the counts the label itself was derived from; (2) `family` and `tool` are weak under every representation and model combination tried (family never exceeds 0.13 macro-F1; tool is additionally unstable across CV splits, ranging 0.03–0.20). This is not obviously a framework or tuning limitation – Section 9's length and channel-identity confound checks rule out two of the more common alternative explanations for weak tabular/text-classification performance, leaving weak-label ceiling (§4.3) and genuine class sparsity (Section 5's 18-class family / 7-class tool taxonomies, several classes with single-digit support) as the remaining candidate explanations.

Table 3 reports only the single best (representation, model) cell per task; the full sweep – 5 tasks × 3 representations × 4 trained models × 3 CV protocols, 180 combinations in total (`outputs/experiments/phase1_*.csv`) – is summarized visually in Figure 1, collapsing the representation axis by taking the best of the three for each (task, model, split) cell so the figure stays legible, and given in full, exact numbers for the stratified-random split in Table 4.

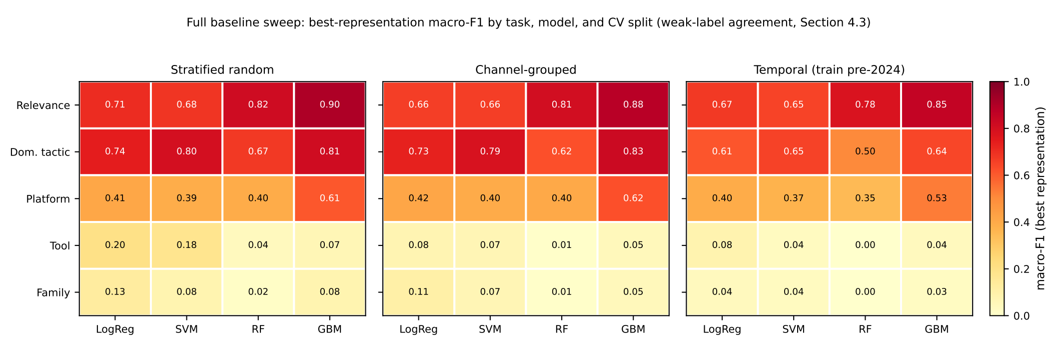

Figure 1: Best-representation macro-F1 for every (task, model) pair, one panel per CV protocol (weak-label agreement, §4.3). `family`/`tool` stay weak under every model and every split; `relevance` is both the strongest task and the most split-robust.

Table 4: Best-representation macro-F1 per (task, model), stratified-random split (weak-label agreement, §4.3). Channel-grouped and temporal splits follow the same pattern with a modest across-the-board drop, per Figure 1.

Task	LogReg	SVM	RF	GBM
Relevance	0.71	0.68	0.82	0.90
Dominant tactic	0.74	0.80	0.67	0.81
Platform	0.41	0.39	0.40	0.61
Tool	0.20	0.18	0.04	0.07
Family	0.13	0.08	0.02	0.08

Figure 2 plots Table 4 as a 3D stem plot rather than solid bars: each (task, model) cell is a thin marked stem rather than an opaque block, so no value is hidden behind a taller neighbor from any viewing angle, which a solid-bar rendering of the same data could not guarantee.

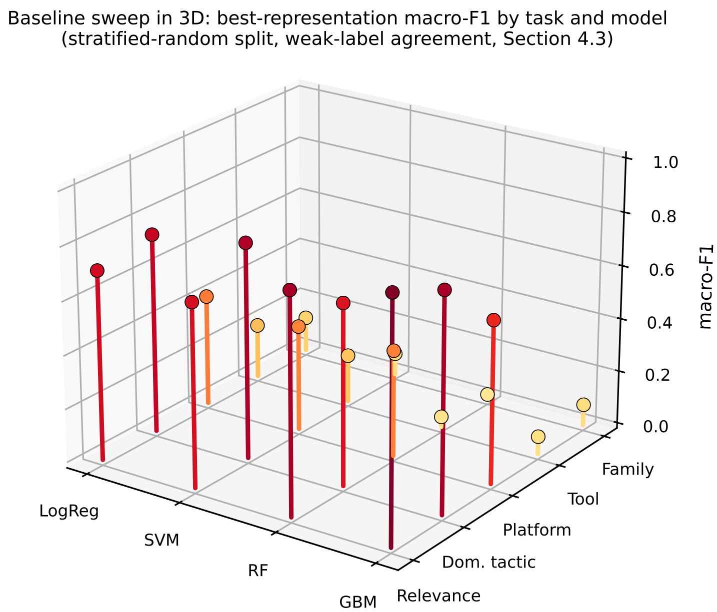

Figure 2: Table 4 as a 3D stem plot – every point visible regardless of viewing angle, unlike a solid-bar rendering of the same data.

6 Uncertainty Quantification

Scope note (§4.3 applies here directly): calibration is normally reported against ground truth – does an 80%-confidence prediction turn out correct roughly 80% of the time. Without independent labels, the only calibration target available here is the weak-label distribution itself, which changes what a well-calibrated result would mean: a model perfectly calibrated against its own training signal has learned to be a confident, well-calibrated echo of a regex, not necessarily a well-calibrated model of ransomware content. This section is retained, with that meaning made explicit at every reported number, because reliability diagrams and Expected Calibration Error against the weak labels are still informative about a narrower question worth answering: does a model's confidence track its *agreement with the extraction pipeline*, or is it uniformly overconfident regardless of agreement. The latter would be a red flag on its own.

6.1 Calibration against weak labels

For a multiclass task, let $\hat{p}(\cdot \mid x)$ be the trained model's predicted class distribution for input x, and let $\hat{c}(x) = \arg\max_c \hat{p}(c \mid x)$ with confidence $\hat{p}_{\max}(x) = \max_c \hat{p}(c \mid x)$. Agreement with the weak label $y^{\mathrm{KA}}(x)$ (the Knowledge Agent's own extraction) is $a(x) = \mathbb{1}[\hat{c}(x) = y^{\mathrm{KA}}(x)]$. For the two multi-label tasks (`family`, `tool`), whose models are independent per-class binary classifiers (Section 5), each (sample, class) pair (x, k) is treated as its own calibration point, with confidence $\hat{p}_{\max}(x, k) = \max\big(\hat{p}_k(x), 1 - \hat{p}_k(x)\big)$ and agreement against the binary weak label thresholded at 0.5.

Reliability is summarized with $B{=}10$ equal-width confidence bins $\{I_b\}_{b=1}^{B}$ and the standard Expected Calibration Error,

$$\mathrm{ECE} \;=\; \sum_{b=1}^{B} \frac{|S_b|}{N} \left|\mathrm{acc}(S_b) - \mathrm{conf}(S_b)\right|, \qquad S_b = \{x : \hat{p}_{\max}(x) \in I_b\}, \tag{1}$$

where $\mathrm{acc}(S_b) = \frac{1}{|S_b|}\sum_{x\in S_b} a(x)$ and $\mathrm{conf}(S_b) = \frac{1}{|S_b|}\sum_{x\in S_b} \hat{p}_{\max}(x)$. Per the scope note above, $a(x)$ measures agreement with y^{KA}, not correctness, so Eq. (1) is a *weak-label ECE*. The headline number below the scope note is the confident-disagreement rate at threshold $\tau = 0.9$,

$$\mathrm{CDR}_\tau \;=\; \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\hat{p}_{\max}(x_i) > \tau] \cdot \big(1 - a(x_i)\big), \tag{2}$$

the closest available analogue to a "confidently wrong" rate without independently verified labels to actually be wrong against.

Each task is evaluated on the 5-fold stratified out-of-fold predictions of its best (feature, model) combination from Table 3, except `family`, whose best combination (embedding representation) needs the dedicated environment described in Section 5's footnote and is reported without a calibration curve here.

Table 5: Weak-label calibration (Equations (1) and (2)) and the RF tree-vote entropy proxy (Equation (3)), best (feature, model) per task. N is the number of (sample) or (sample, class) calibration points. Entropy AUROC is how well entropy alone separates agreement from disagreement with the weak label (Algorithm 1).

Task	N	ECE	$\mathrm{CDR}_{0.9}$	Entropy (agree/disagree)	Entropy AUROC
Relevance	1034	0.079	0.071	0.910 / 0.980	0.769
Dominant tactic	1034	0.051	0.043	0.275 / 0.551	0.860
Platform	1034	0.110	0.104	0.717 / 0.769	0.675
Tool (per-label)	7238	0.016	0.016	0.121 / 0.271	0.977
Family (per-label)	1034	–	–	0.095 / –	–

Figure 3 plots the reliability curves underlying Table 5. Two things worth naming honestly rather than smoothing over: first, per-bin counts are small (single digits to low tens per bin at N=1034), so individual bin heights are noisy – the curves are directionally informative, not point-estimate-precise. Second, `family`'s row has no ECE/CDR: its calibration point is unavailable in this environment (as above), so only the RF tree-vote entropy proxy, computed on the TF-IDF fallback representation, is reported for it, and even that proxy has no "disagree" entries to average – at the per-(sample, class) level, most of `family`'s 18 classes are negative for most samples, so the majority-class prediction ("absent") agrees with the weak label on essentially every RF-on-TF-IDF calibration point despite the task's low macro-F1 in Table 3, which weights all 18 classes equally rather than by prevalence. Both readings are correct simultaneously; they are just answering different questions (per-pair agreement vs. per-class balanced performance).

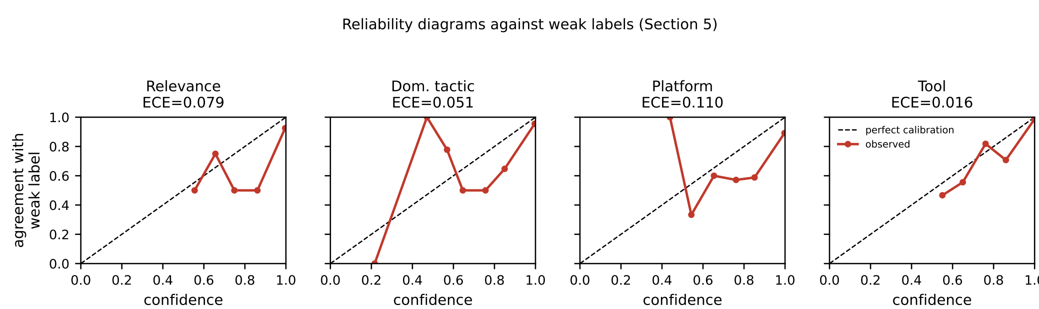

Figure 3: Reliability diagrams against weak labels, one panel per task with a usable calibration curve (Table 5). Diagonal is perfect calibration.

6.2 A free uncertainty proxy: Random Forest tree-vote entropy

Reliability diagrams need a held-out calibration protocol; a Random Forest offers a per-prediction uncertainty signal with no separate calibration step. For an input x and a forest of T trees producing individual (hard) predictions $t_1(x), \ldots, t_T(x)$ over K classes, let $\pi_c(x) = \frac{1}{T}\sum_{j=1}^{T} \mathbb{1}[t_j(x) = c]$ be the empirical vote share for class c. The normalized tree-vote entropy is

$$H(x) \;=\; -\frac{1}{\log K}\sum_{c=1}^{K} \pi_c(x)\, \log \pi_c(x) \;\in\; [0,1], \tag{3}$$

with $H(x)$=0 when every tree agrees and $H(x)$=1 at a uniform split across all K classes. For the multi-label tasks, K=2 per class and $H(x, k)$ is averaged over classes k for a per-sample summary. Algorithm 1 is the exact out-of-fold procedure used to produce Table 5's entropy columns.

Algorithm 1 RF tree-vote entropy, 5-fold out-of-fold

Require: dataset $\mathcal{D}$, task with label function $y(\cdot)$, feature builder ϕ

$\mathcal{F} \leftarrow 5$ stratified folds of $\mathcal{D}$
for fold $(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) \in \mathcal{F}$ **do**
    fit ϕ on $\mathcal{D}_{\text{train}}$; fit Random Forest $\{t_j\}_{j=1}^{T}$ on $\phi(\mathcal{D}_{\text{train}})$
    for $x \in \mathcal{D}_{\text{test}}$ **do**
        $t_j(x) \leftarrow$ individual tree prediction, $j = 1, \ldots, T$ ▷ sklearn returns internal integer-encoded classes here, not the original label strings – see note below
        recover original class labels via the fitted forest's class index before comparing to $y(x)$
        compute $H(x)$ via Equation (3); record $a(x) = \mathbb{1}[\text{forest majority vote} = y(x)]$
    end for
end for
return $\{(H(x), a(x))\}_{x\in\mathcal{D}}$, pooled across folds

The in-line comment in Algorithm 1 records a real implementation pitfall, not a hypothetical one: scikit-learn fits each tree in a `RandomForestClassifier` on internally label-encoded integer targets, so `tree.predict()` returns class *indices*, not the original label strings the ensemble's own `.predict()` returns. Comparing tree-level votes directly against string labels without mapping back through the fitted forest's class index silently produces 0% apparent agreement on every multiclass task – caught in this project's own Phase 2 run (initial numbers showed exactly that pattern across all three multiclass tasks simultaneously, which is what made it legible as a bug rather than a real result) and is recorded here as a concrete instance of the caution Section 10 urges about verifying rather than assuming standard library internals. Table 5's entropy AUROC column shows the proxy works well once fixed: 0.68–0.98 across tasks, best on `tool` (many easy true negatives) and weakest on `platform`.

7 Conformal Prediction

Scope note: split conformal prediction's distribution-free coverage guarantee is a statement about exchangeability with the calibration set, not about label correctness – it holds formally even when calibrated against weak labels, because the guarantee is "the true *weak* label falls in the predicted set $(1-\alpha)$ of the time," not "the true *real-world* label does." That distinction is the entire content of this scope note: conformal coverage against the Knowledge Agent's labels is a well-defined, honestly reportable number, but it validates agreement with the extraction pipeline, not correctness. Reported as such.

7.1 Method: split conformal with the LAC score

Split conformal prediction (Vovk et al., 2005, Lei et al., 2018; see Angelopoulos and Bates, 2023 for a tutorial treatment) uses a nonconformity score $s(x, y)$ that is small when y is a plausible label for x. This paper uses the least-ambiguous-set (LAC) score,

$$s(x, y) \;=\; 1 - \hat{p}(y \mid x), \tag{4}$$

calibrated on a held-out split $\mathcal{D}_{\text{calib}}$ of size n disjoint from both the training and test data. The calibration threshold uses the standard finite-sample correction,

$$\hat{q} \;=\; \text{Quantile}\left(\{s(x_i, y_i)\}_{i=1}^{n}\,;\; \frac{\lceil (n+1)(1-\alpha)\rceil}{n}\right), \tag{5}$$

and the prediction set for a new input x is

$$C(x) \;=\; \{\, y \;:\; \hat{p}(y \mid x) \geq 1 - \hat{q} \,\}. \tag{6}$$

Under exchangeability of $\mathcal{D}_{\mathrm{calib}}$ and the test point, this construction guarantees

$$\mathbb{P}[\, y_{\mathrm{test}} \in C(x_{\mathrm{test}}) \,] \;\geq\; 1 - \alpha, \tag{7}$$

with α=0.10 (90% target coverage) throughout. As the scope note above states, y_{test} here is the weak label y^{KA}: Equation (7) is a real, distribution-free guarantee about agreement with the extraction pipeline, not about real-world correctness. For `dominant_tactic` and `platform`, Equations (4) to (6) apply directly over the task's full class vocabulary. `family` and `tool` have no single multiclass structure to conformalize – each is an independent per-class binary classifier (Section 5) – so each class k gets its own binary LAC calibration and its own $\hat{q}_k$, and a sample's per-label prediction sets are reported both individually and jointly (all labels simultaneously covered).

Algorithm 2 is run for 5 repeated random 60/20/20 train/calibration/test splits per task, results averaged across repeats rather than reported from one arbitrary split – a single 20% calibration set at this sample size (n≈200) is not a reliable enough estimate of $\hat{q}$ to report alone.

Algorithm 2 Split conformal calibration and coverage evaluation

Require: dataset $\mathcal{D}$, task, target coverage $1 - \alpha$, repeats R
for $r = 1, \ldots, R$ **do**
  split $\mathcal{D}$ into $\mathcal{D}_{\mathrm{train}}$ (60%), $\mathcal{D}_{\mathrm{calib}}$ (20%), $\mathcal{D}_{\mathrm{test}}$ (20%), stratified
  fit model on $\mathcal{D}_{\mathrm{train}}$
  compute $\hat{q}$ on $\mathcal{D}_{\mathrm{calib}}$ via Equations (4) and (5)
  for $x \in \mathcal{D}_{\mathrm{test}}$ **do**
    build $C(x)$ via Equation (6); record $\mathbb{1}[y^{\mathrm{KA}}(x) \in C(x)]$ and $|C(x)|$
  end for
end for
pool test-set coverage indicators across all R repeats
report marginal coverage, and coverage restricted to the Year≥2026 and Conference-channel slices

7.2 Results

Table 6: Split-conformal coverage at 90% target (α=0.10), pooled over 5 repeats, n≈1035 pooled test predictions per task. `family`/`tool` report mean per-label coverage and the stricter all-labels-jointly-covered rate. Set size is out of the task's class count (K) for `dominant_tactic`/`platform`, or out of 2 (per label) for `family`/`tool`.

Task	K	Coverage (marginal)	Avg. set size	Slice: Yr≥26	Slice: Conf.
Dominant tactic	11	0.877	0.94/11	0.890	0.870
Platform	4	0.925	1.19/4	0.932	0.899
Family (per-label)	18	0.912 (joint: 0.446)	0.93/2	0.487^{joint}	0.572^{joint}
Tool (per-label)	7	0.909 (joint: 0.650)	0.92/2	0.640^{joint}	0.752^{joint}

Figure 4 plots Table 6 directly. Reading the results against the two questions this section set out to answer: first, does the guarantee hold empirically – `dominant_tactic` and `platform`'s marginal coverage (0.877, 0.925) both sit close to the 0.90 target, with set sizes well under one class on average (0.94/11, 1.19/4), i.e. the conformal sets are usually a single class or empty, not the whole vocabulary hedged defensively. Second, the conditional-coverage stress test: neither the Year≥2026 nor the Conference-channel slice shows the kind of collapse that would flag a real distribution-shift problem for these two tasks (all four slice numbers stay within about 3 points of the marginal rate). `family` and `tool`'s per-label coverage (0.91, 0.91) is equally close to target, but their *joint* (all-labels-simultaneously) coverage is far lower (0.45, 0.65) – an expected consequence of compounding per-label error rates across 18 and 7 largely-independent binary calibrations respectively, not a failure of Equation (7) (which was never a joint guarantee to begin with) and not itself evidence of conditional-coverage collapse, since the joint-coverage slices track the joint-coverage marginal about as closely as the multiclass tasks' slices track theirs. Per Section 7.1, all of the above describes agreement with y^{KA}, and Section 8 below is the complementary analysis that does not carry that caveat.

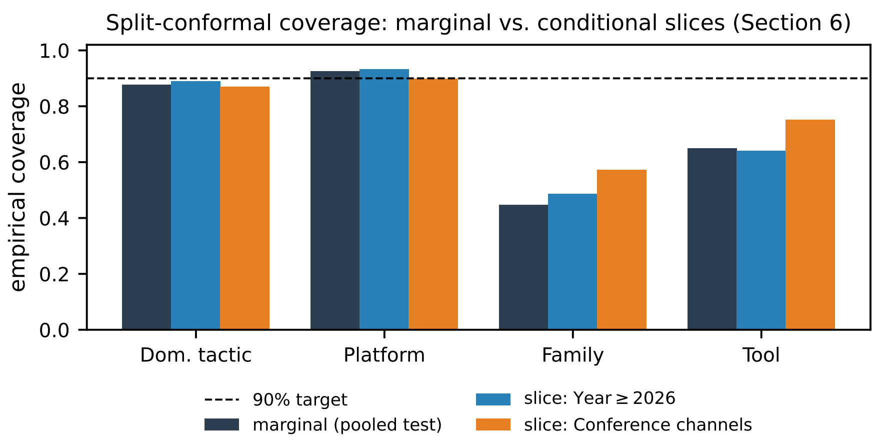

Figure 4: Empirical conformal coverage, marginal vs. the two conditional-coverage stress-test slices, against the 90% target (Table 6).

8 Out-of-Distribution Detection

Unlike Sections 5–7, this section's results are not subject to the weak-label scope note in Section 4.3: OOD detection asks whether a model's own confidence or embedding-space behavior differs between in-distribution and out-of-distribution input, which is answerable without knowing whether either input's *label* is correct.

8.1 Setups and methods

Four setups, each with a known-by-construction in-distribution (ID)/out-of-distribution (OOD) split, drawn from real structure in the corpus rather than synthetic corruption:

1. **Temporal emergence.** ID: pre-2024 videos. OOD: post-2024 videos mentioning a family whose first appearance in this corpus is 2024 or later – verified directly against per-family first-occurrence year rather than assumed (Maze, Medusa, Black Basta, INC: 2024; Cactus: 2025), at Algorithm 3's assertion step.
2. **Channel-type shift.** ID: Vendor + DFIR channels. OOD: Conference + Independent channels.
3. **Leave-rare-family-out.** ID: videos never mentioning "Cl0p" (11 occurrences total – long-tailed, unlike Conti/LockBit/ Qilin/Akira, each with 19+). OOD: the 11 Cl0p-mentioning videos.
4. **External corpus.** ID: RansKnow transcripts. OOD: 20 Newsgroups (`rec.sport.baseball + sci.space`), included as a sanity check that the methods below can detect an obviously unrelated domain before trusting them on setups 1–3.

Four detection methods, computed from a shared `dominant_tactic` classifier (structured features, LightGBM – Table 3's strongest non-circular result, and a natural multiclass fit for the softmax-family methods below), refit fresh on each setup's ID-only training pool so that no OOD signal leaks into fitting:

$$s_{\mathrm{MSP}}(x) = 1 - \max_c \hat{p}(c \mid x) \tag{8}$$

$$s_{\mathrm{energy}}(x) = -\log \sum_c \exp\big(f_c(x)\big) \tag{9}$$

$$s_{\mathrm{maha}}(x) = \min_c \big(\phi(x) - \mu_c\big)^{\top} \Sigma^{-1} \big(\phi(x) - \mu_c\big) \tag{10}$$

$$s_{\mathrm{conf}}(x) = 1 - \hat{p}\big(\hat{c}(x) \mid x\big) \qquad \text{(Equation (4) at the predicted class)} \tag{11}$$

Equation (8) is Hendrycks and Gimpel's max-softmax-probability baseline [Hendrycks and Gimpel, 2017]. Equation (9) is the energy score [Liu et al., 2020], computed from LightGBM's raw pre-softmax margins $f_c(x)$ (*not* from $\hat{p}(c \mid x)$: probabilities already sum to 1, which makes Equation (9) degenerate to a constant if computed from them directly – a real pitfall caught while implementing this section, not a hypothetical one). Equation (10) is the class-conditional Gaussian Mahalanobis distance of Lee et al. [2018], with $\phi(x)$ the 384-dim mean-pooled sentence embedding (Section 5's footnote) and μ_c, Σ the ID-train class-conditional means and pooled covariance, using the classifier's own *predicted* classes to form groups (consistent with Section 4.3: no independently verified labels exist to group by instead). Equation (11) reuses Section 7's LAC score, evaluated at the predicted rather than true class for OOD flagging – this is deliberately, not accidentally, the same quantity as Equation (8); the two are reported separately for comparability with Equation (5)'s calibrated threshold (does a point's own score exceed the 90%-target $\hat{q}$), which is the operationally distinct number.

Algorithm 3 OOD setup evaluation

Require: corpus $\mathcal{D}$, ID/OOD partition rule for the setup
1: **assert** every claimed-emerging family's first-occurrence year matches the setup's stated cutoff ▷ caught a factual error at this step – see note below
2: split ID pool into $\mathcal{D}_{\text{train}}$ (75%) / $\mathcal{D}^{\text{ID}}_{\text{test}}$ (25%)
3: fit `dominant_tactic`/structured/LightGBM on $\mathcal{D}_{\text{train}}$
4: score $\mathcal{D}^{\text{ID}}_{\text{test}}$ and $\mathcal{D}_{\text{OOD}}$ under Equations (8) to (11)
5: report AUROC(ID vs. OOD) per method, plus the conformal exceedance-rate contrast

The assertion in Algorithm 3 is not decorative: the temporal- emergence setup's family list was originally drafted from memory as "Qilin, DragonForce, Black Basta," before checking actual first-occurrence years against the corpus. Qilin's first real occurrence is 2019, not post-2023, and DragonForce does not appear in this corpus at all – both would have silently produced a meaningless setup (there is no real ID/OOD separation if the "emerging" families already appear throughout the ID training period). The corrected list (Maze, Medusa, Black Basta, INC, Cactus) is the one actually used in Table 7 and is verified by Algorithm 3 on every run, not just checked once by hand.

8.2 Results

Table 7: OOD-detection AUROC by setup and method (Equations (8) to (11)). $n_{\mathrm{ID}}/n_{\mathrm{OOD}}$ are ID-test/OOD sample counts. Mahalanobis is unavailable for `external_corpus` (20 Newsgroups has no entry in the RansKnow-only embedding cache).

Setup	n_{ID}	n_{OOD}	MSP	Energy	Mahalanobis	Conformal
Temporal emergence	55	31	0.561	0.360	0.223	0.533
Channel-type shift	156	353	0.477	0.487	0.666	0.487
Leave-rare-family-out	256	11	0.377	0.350	0.145	0.376
External corpus (sanity)	259	1190	0.625	0.476	–	0.631

Figure 5 plots Table 7. The honest reading is a mixed and mostly negative one, which is itself the finding worth reporting rather than a result to explain away. Three observations:

The confidence-based methods (MSP, energy, conformal) hover near chance (0.5) on every setup, including the sanity check. Diagnosing *why* on `external_corpus` is informative: 99.6% of

20 Newsgroups documents are classified `None` (no dominant tactic detected) by the shared classifier, and with *higher* average confidence than the real in-distribution test set (mean $\hat{p}_{\max}$: 0.921 OOD vs. 0.895 ID). A near-empty, off-topic TF-IDF vector and a genuinely tactic-free ransomware transcript both produce the same signature – confidently "no tactics detected" – so a classifier whose decision function is fundamentally an argmax over small keyword-mention counts has no way to distinguish "nothing here" from "wrong domain entirely." This is a known blind spot of softmax/probability-based OOD detection in general [Hendrycks and Gimpel, 2017], reproduced concretely here rather than assumed from the literature.

Mahalanobis distance on real sentence embeddings is the one method that separates classes at all, and only partially. It reaches 0.666 on channel-type shift – Conference/Independent channels differ enough in embedding space (production style, framing, audience) to be mildly detectable – but performs *worse* than chance on temporal emergence (0.223) and leave-rare-family-out (0.145). Both of those setups' OOD population describes the same underlying phenomenon (ransomware incidents, attacker tradecraft) as the ID population, just naming a different or newer group – exactly the case where sentence embeddings, which encode semantic content rather than proper nouns, would *not* be expected to separate the two populations well. The small OOD sample sizes in these two setups (n=31, n=11) also mean each AUROC estimate carries substantial variance on its own; with n_{OOD}=11, a single re-ordered pair moves the estimate by roughly $1/11$.

The conformal exceedance-rate contrast (Equation (5)'s $\hat{q}$) is uninformative in this corpus specifically. In every setup, the calibrated threshold $\hat{q}$ lands so close to the maximum possible score ($\hat{q} \geq 0.99$ in three of four setups) that essentially no point, ID or OOD, exceeds it – a consequence of the classifier being very confidently correct on the bulk of its small ($n{\approx}115$) calibration split, which pushes the 90th-percentile score to an extreme value. Equation (11)'s AUROC column is the more informative number from this method here; the exceedance-rate contrast is reported in the released JSON (`outputs/phase4_ood.json`) for completeness but is not a useful discriminator at this sample size.

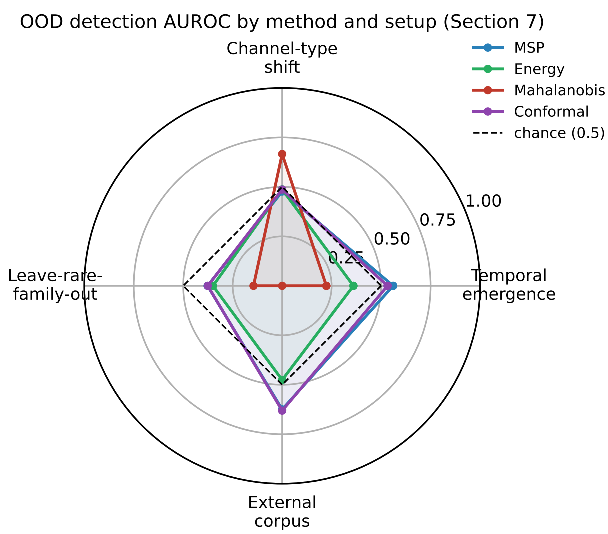

Figure 5: OOD-detection AUROC by method (color) and setup (axis), against the chance line at 0.5 (Table 7). Mahalanobis is the only method to clear chance by a wide margin, and only on one of four setups.

Taken together, Section 8.2 does not support a general claim that this pipeline's models reliably detect distributional shift; it supports a narrower, still useful one: OOD detection here is hard, softmax-confidence-based methods have an identifiable and reproducible blind spot on this specific classifier (the "no signal" vs. "wrong domain" confusion above), and embedding-distance methods help only when the shift is stylistic rather than topical. Unlike Sections 5 to 7, none of this depends on whether y^{KA} is correct (Section 4.3) – it is a direct, if modest, empirical finding about model behavior.

9 Interpretability and Perturbation Robustness

Unlike Sections 5–7, this section asks questions about model *behavior* – does it exploit shortcuts, what does it rely on, how much do its predictions move under semantically-irrelevant edits – none of which require independent ground truth to answer, so Section 4.3's scope note does not apply here.

9.1 Confound checks

Length. Transcript word count correlates moderately with `Tactic_Total_Mentions` (Pearson $r = 0.409$) and weakly with `Family_Count` ($r = 0.131$), but not at all with `Tool_Total_Mentions` ($r = 0.043$, not significant) – the confound is not uniform across signal types. Because it is mathematically guaranteed that dividing every tactic's count for one video by that video's own word count cannot change which tactic has the highest count (argmax is scale-invariant per row), the confound cannot affect how the `Dominant_Tactic` label itself was derived; the question that matters is whether it inflates `dominant_tactic`'s strongest result (structured features, Table 3). Re-scoring with length-normalized (mentions per 1,000 words) instead of raw-count features gives 0.753 macro-F1 against the raw-count model's 0.764 – a 1.1-point difference, not a length-exploitation artifact.

Channel identity. 486 of 1,034 transcripts (47%) contain at least one literal mention of their own channel's name. Masking those mentions and re-scoring TF-IDF LogReg under both CV protocols leaves the `channel_grouped-vs-stratified_random` gap unchanged for `family` (0.028 raw → 0.031 masked) and for `dominant_tactic` (0.020 raw → 0.029 masked). Whatever a model exploits when a channel is not held out, it is not the channel literally naming itself – the leakage this split protocol exists to catch is stylistic or topical, not addressable by name-masking.

9.2 Counterfactual explanations

Motivated directly by Finding A: does a classifier trained on the *fixed* labels still show the regex's original word-collision fragility? A TF-IDF+LogReg model was trained on today's corrected `Family` labels and evaluated on the 275 videos Section 4.1 confirmed as false positives under the pre-fix regex (videos where "`play`" appears without ransomware context). The trained model assigns essentially zero probability to `Play` on all 275 (maximum $P = 0.013$) – it did not inherit the bare-word-collision bug.

The complementary question – what drives the model's actual positive `Play` predictions – was then asked with a greedy word-*type* removal counterfactual search (remove the vocabulary word whose removal most reduces $P(\text{Play})$, repeat until the prediction flips or a budget is exhausted) on the 8 highest-confidence true positives. `play` appears in 5 of 8 removal sets and is consistently one of the highest-impact single removals when present, but **0 of 8 cases flip on removing "play" alone** – every flip required 4–10 additional removals, most of them generic (function words, not ransomware-specific vocabulary). The model reduced, but did not eliminate, reliance on the single ambiguous trigger word, and the words it combines that reliance with are not obviously a clean, interpretable "ransomware vocabulary" set – worth flagging as its own finding about decision-boundary diffuseness, independent of the word-collision question it was designed to answer.

9.3 Perturbation robustness

Untargeted counterpart to the counterfactual search above: two perturbation types, measuring flip rate (does the top prediction change) and mean absolute probability drift on freshly-fit TF-IDF+LogReg models. *Simulated ASR noise* (compound proper nouns split apart, e.g. `WannaCry`→"wanna cry," `Mimikatz`→"mimi cats," plus 8% random token deletion) was applied to the 74 official-caption videos specifically – the "clean" subset, simulating what those transcripts would look like if they had gone through the ASR cascade that produced the other 92.7% of the corpus (Finding B). *Synonym substitution* (common non-jargon words swapped for synonyms) was applied to a 150-video sample.

`dominant_tactic`, `platform`, and `relevance` are essentially unmoved by either perturbation (0.0–0.7% flip rate, mean drift < 0.01 in every case). `family` flips 20–26% of the time under both perturbations despite a much smaller mean drift ($\sim$0.002) than any other task – not a new problem, but the same weak-model issue from Table 3 (`family`'s best macro-F1 is 0.129) surfacing differently: closely-bunched, low-confidence probabilities mean a negligible nudge is enough to change which barely-ahead class wins. On the framing of "adversarial" robustness for this dataset: no real adversary submits transcripts to evade this classifier in production, so a full gradient-based adversarial-attack framework would be disproportionate to what this paper needs; the actual risk surface is ASR transcription noise and channel-style variation, which is what this sweep directly measures. Combined with Section 8's temporal-emergence and leave-rare-family-out setups, these results are the paper's answer to how the system handles evolving, novel, and adversarial-adjacent input – not three separate open questions.

10 Limitations and Ethics

This paper's single most consequential limitation is the absence of independent ground truth, discussed at length below; it opens this section as prose, not because the remaining limitations are unimportant, but because they are all of a smaller and more familiar kind (sampling bias, ASR noise, dual use) that most empirical NLP papers carry in some form, whereas the ground-truth gap changes what every quantitative claim in Sections 5–7 *means*, not merely how confidently it can be stated.

- **No independent ground truth.** This is the paper's central limitation, not one bullet among several, and is why it is stated repeatedly rather than once: this project does not include a human-annotated evaluation set independent of the Knowledge Agent pipeline (Section 4.3). External annotator recruitment required institutional ethics approval not available to this project; a single-investigator annotation pass was considered and rejected as still too close to the system under evaluation to count as independent evidence. Every result in Sections 5–7 is consequently agreement with an audited but demonstrably imperfect extraction pipeline (Finding A), not verified accuracy – Section 8's OOD results and Section 9's behavioral diagnostics are the exceptions, since neither depends on label correctness. Recruiting independent annotators, whether through formal ethics approval or an unpaid collaborator arrangement that falls outside whatever definition of human-subjects research applies at the author's institution, is the single highest-value extension of this work – more valuable than any additional modeling in Sections 5–7 would be, since none of that modeling can currently be checked against anything but itself.
- **Sampling bias.** The corpus is conditioned on which channels operators chose to run, which videos they chose to title and describe in keyword-matchable ways, and YouTube's own recommendation and search surfacing – not a random or exhaustive sample of ransomware-relevant discourse.
- **ASR noise.** 92.7% of transcripts are machine-transcribed, mostly by `whisper_base`. Section 9.3's simulated-ASR-noise sweep bounds how much this moves predictions; Finding B's word-error-rate-on-jargon question (comparing ASR output against source audio directly, rather than a simulation of ASR-style errors) remains open.
- **Weak-label ceiling.** Every quantitative result in Sections 5–7 is bounded by the Knowledge Agent's own precision/recall, which Finding A shows is not perfect even after the disambiguation fix, and which Section 9.2 shows partially propagated into at least one trained model.
- **Dual use.** This is a corpus of ransomware tactics, tools, and case studies extracted from public educational content, and it is worth being direct about what that does and does not change. Every underlying video is already public, already indexed by YouTube's own search, and produced by defenders (vendors, DFIR practitioners, conference speakers) specifically to be watched by other defenders; RansKnow discloses no technique, tool, or tactic that is not already public, and the source material is materially less operationally detailed than existing public resources built for exactly this kind of aggregation – the MITRE ATT&CK knowledge base itself, and the CTI reporting datasets in Section 2 [Lim et al., 2017, Lange et al., 2024], both already structure comparable tactic/technique/tool information at finer granularity and higher label quality than RansKnow's weak-label extraction achieves (Section 4.3). What RansKnow adds is not new operational detail but a different modality (transcribed spoken narration rather than written reports) and different scale accessible without specialized CTI tooling. The realistic marginal risk is a small reduction in the effort needed to *aggregate* already-scattered public information into one structured, searchable place, rather than any

disclosure of non-public technique detail; this is the same tradeoff every public threat-intelligence aggregation resource makes; and the corpus's actual utility, given Section 4.3's weak-label ceiling and Section 5's weak family/tool classification performance, currently favors defensive research uses (weak-supervision methodology, corpus-level trend analysis) far more than it favors any operational offensive use, for which primary-source technical documentation would be a far more direct and reliable starting point than this corpus's noisy, ASR-transcribed, regex-extracted labels.

11 Conclusion

RansKnow is a 1,128-video, 1,034-transcript, MITRE ATT&CK-aligned corpus of ransomware-related YouTube content, and this paper's contribution is not the corpus alone. The label-quality audit methodology (Section 4) and the interpretability diagnostics that verify it against a trained model's actual behavior (Section 9) are a second, reusable contribution: any weak-label security-NLP dataset built the same way (keyword or regex extraction over scraped or transcribed text) can reuse the alias-collision-audit methodology and the counterfactual/ perturbation checks even without reusing this specific corpus, and should – Finding A demonstrates concretely that a plausible-looking regex label can be dominated by false positives (Table 1: 78% of pre-fix `Play` detections), which is precisely the class of error a corpus released without this kind of audit would carry silently.

The most valuable open follow-up to this specific paper is not further modeling. It is independent, human-annotated verification of a subset of RansKnow's labels (Section 10), and it is worth being explicit about why that ranks above any modeling extension: every quantitative result this paper reports outside Section 8 would change in *meaning*, not just in number, the day that verification exists. A macro-F1 of 0.813 on `dominant_tactic` currently means "a LightGBM classifier reproduces the Knowledge Agent's own regex output at this rate"; the same number, computed against even a modest independently-annotated subset, would mean something closer to what a reader of an ML paper's results table normally assumes a reported score means. Closing that gap changes the paper's central claim, not just its confidence interval, which is why it is listed first among follow-ups rather than alongside ordinary future-work suggestions like additional model architectures or task definitions.

Acknowledgments and Disclosure of Funding

This work was supported in part by the Pro:NE (Pronortheast) Network under Grant RDF22/proNE/JHA. The authors thank the members of the CyberNets Research Group, Northumbria University, for their insightful discussions and useful feedback throughout this research project. The opinions, findings, conclusions, or recommendations expressed in this material are solely those of the authors and do not necessarily represent the views of the supporting team. The authors declare no competing interests.

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A Keyword List

The 21-term inclusion keyword list, matched case-insensitively against video title and description (Section 3), verbatim from `Scripts/fetch_transcripts.py`'s `KEYWORDS` constant:

```
ransomware, extortion, double extortion, encrypt, encryption,
decrypt, data leak, incident response, ‘‘ir’’, lockbit, alphv,
blackcat, conti, ryuk, cl0p, clop, akira, revil, sodinokibi,
ransomware-as-a-service, raas
```

The bare token "`ir`" is matched with surrounding spaces (`" ir "`) specifically to avoid matching it as a substring of unrelated words.

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   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 10 opens with "No independent ground truth" as the paper's central limitation rather than a minor bullet, and covers sampling bias, ASR transcription noise, the weak-label ceiling, and dual-use considerations.

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Answer: [N/A]

Justification: The paper contains no theoretical results or proofs – it is an empirical dataset and baseline-framework contribution.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 specifies the channel/video selection criteria and full keyword list (Appendix A); Section 5 specifies the task taxonomy, feature representations, models, and CV protocols. All extraction, audit, and experiment code is in the public repository (see next item), including the exact hyperparameters used (default scikit-learn/LightGBM settings unless otherwise noted in code).

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5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Dataset: https://www.kaggle.com/datasets/henrykabuye/ransknow-v1 (CC0-1.0). Code: https://github.com/Henrinnes/RansKnow (fetch pipeline, label-quality audit scripts, `rk_pipeline` experiment framework, Phase 5 confound/counterfactual/perturbation scripts), released under the MIT License (`LICENSE` in the repository root).

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6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 5 specifies the three CV protocols (stratified random, channel-grouped, temporal) and the five-model/three-representation grid; exact hyperparameters (regularization, tree counts, TF-IDF vocabulary size, embedding model and chunking) are documented in `Scripts/rk_pipeline/` and referenced in Section 5.

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7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Reported macro-F1 values are single-seed, 5-fold (or single-split, for the temporal protocol) point estimates, not averaged or bounded across repeated random seeds.

[TODO] This is a real, addressable gap, not an inherent limitation – the framework's split protocols already accept a seed parameter; re-running the Phase 1 grid across 3–5 seeds and reporting standard deviation is worth doing before submission, given how cheap most combinations were to run (see compute-resources item below).

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8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Experiments ran on a local workstation: 32-core CPU (structured/TF-IDF representations, all sklearn/LightGBM models) plus 3×NVIDIA GTX 1070 Ti, 8GB each (embedding representation, sentence-transformer encoding). Observed wall-clock time: individual task/model/split combinations ranged from seconds to ∼80s for the heaviest (RandomForest on 18-class multilabel TF-IDF); full grids (5 tasks × up to 5 models × 3 splits) completed in 20–45 minutes per representation.

   [TODO] State total compute across the whole project, including exploratory runs that didn't make the paper, per the guideline – not tracked precisely during development.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

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Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

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